

Topology-Synchronous Decoupled C–ESR Identification for the Energy-Transfer Capacitor in Ćuk Converters

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Abstract—The energy-transfer capacitor in a Ćuk converter is both a power-transfer element and a reliability-sensitive component, but ordinary ripple measurements couple its capacitance and equivalent series resistance (ESR). This paper develops physically direction-decoupled, topology-synchronous observations that exploit the inherent current commutation between $+i_{L1}$ and $-i_{L2}$ without diagnostic injection. Timestamped fits in safe pre- and post-edge windows reconstruct an ESR-dominant voltage discontinuity, whereas in-state charge integration yields a capacitance-dominant row after ESR compensation. Positive finite-window information bounds establish local identifiability under valid continuous-conduction-mode conditions. A topology-synchronous direction-specific recursive least-squares estimator (TS-D-RLS) provides the primary lightweight realization; a structured linear time-varying Kalman estimator (TS-SLTVKE) is retained for uncertainty-aware reporting. On 48 blind operating/health cases evaluated under four noise profiles and four residual timing offsets, TS-D-RLS achieved mean capacitance and ESR absolute percentage errors of 0.3727% and 0.2233% and a mean internal convergence time of 13.21 switching cycles. It also gave the better aggregate tracking tradeoff over 0.1–100 s degradation ramps, while TS-SLTVKE produced better joint interval coverage. A device-realistic TMS320F28379D simulation yielded worst-case 95th-percentile capacitance and ESR errors of 1.6425% and 2.6109%. The results support the observation design and constrained embedded feasibility within simulation; hardware and accelerated-aging validation remain open.

Index Terms—Ćuk converter, capacitor condition monitoring, energy-transfer capacitor, capacitance estimation, equivalent series resistance, parameter identification, recursive least squares, power electronics reliability.

I. INTRODUCTION

Capacitance loss and equivalent series resistance (ESR) growth alter electrical stress, thermal loss, and filtering performance in power converters. Online monitoring methods therefore span ripple analysis, injected excitation, circuit re-configuration, observers, recursive identification, and data-driven processing [1], [2]. Their applicability depends on the capacitor’s circuit role and on which electrical variables are already measured.

The energy-transfer capacitor in a Ćuk converter is distinct from a dc-link or output filter: it lies in the main power-transfer path and alternately carries $+i_{L1}$ and $-i_{L2}$. Normal switching thus creates a bidirectional current commutation that can expose capacitor health without an added diagnostic perturbation. Treating this component as a generic two-terminal capacitor sampled asynchronously discards that topology-specific information.

Prior joint C–ESR methods use inherent ripple or within-cycle relations [3]–[5], switched networks and observers [6]–[8], or inherent and injected spectral features [9]–[11]. Related recent work includes sensorless Boost ESR estimation, optimized Buck-capacitor injection, and large-signal C–ESR estimation [12]–[14]. Wavelet/Kalman and recursive RLS/Kalman formulations provide further signal-processing and estimation routes [15]–[18]. Ćuk-specific modeling, fault diagnosis, and prognostics have also been studied [19]–[22]. These studies establish the broader tools; the gap addressed here is how the Ćuk transfer mechanism itself can be separated into implementable edge and charge information.

This paper reconstructs the switching-edge discontinuity for an ESR-dominant observation and uses an ESR-compensated in-state charge balance for a capacitance-dominant observation. Table I positions that scope against representative methods. The comparison is categorical because the converter topologies, sensing sets, and reporting protocols differ.

The contributions are fourfold.

- 1) The normal Ćuk switching sequence is interpreted as topology-induced health excitation through the $+i_{L1} \leftrightarrow -i_{L2}$ capacitor-current commutation.
- 2) Timestamp-reconstructed edge and safe-window charge observations provide ESR-dominant and capacitance-dominant information while preserving the distinction between terminal and internal capacitor voltage.
- 3) Positive finite-window information bounds establish local identifiability under valid continuous-conduction-mode conditions. TS-D-RLS is the primary low-cost realization, while TS-SLTVKE is an uncertainty-aware extension; the standard RLS and Kalman recursions are not claimed as novelties.
- 4) A common evaluation protocol combines independent plant traces, 48 blind cases, noise/timing sweeps, an observation–estimator factorial, degradation trajectories, operating transients, and a TMS320F28379D device-realistic simulation.

The remainder of the paper develops the model and observations, proves the finite-window bounds, presents the two realizations, and evaluates their accuracy, dynamics, uncertainty, and embedded feasibility.

TABLE I
REPRESENTATIVE RELATED METHODS AND THE SCOPE OF THIS WORK

Method	Capacitor role	Joint C/ESR	Added excitation	Measurements	Processing	Distinction from this work
Hasegawa <i>et al.</i> [3]	inverter dc link	yes	no	dc-link voltage and ripple current	ripple-frequency separation	different topology and ripple bands
Yao <i>et al.</i> [4]	Buck output filter	yes	no	PWM timing and output voltage	within-cycle relation	topology-specific relation without Ćuk current reversal
Barbulescu <i>et al.</i> [6]	device under test	yes	switched network	terminal voltage and switched resistive network	parameter observer	requires an additional discharge network
Ribeiro <i>et al.</i> [9]	NPC dc link	yes	no	converter voltage/current signals	spectral features plus RLS	different inherent-signal mapping
Amaral <i>et al.</i> [15]	boost output/dc link	yes	no	converter signals	wavelet plus Kalman filtering	reconstructed frequency features rather than edge/charge rows
This work	Ćuk energy transfer	yes	no	v_T , i_1 , i_2 , PWM state/time	edge fit and charge balance	uses $+i_{L1} \leftrightarrow -i_{L2}$ commutation

Sensing note: No diagnostic injection does not imply zero sensing burden. The device-realistic realization uses separate absolute- and edge-voltage ranges in the differential v_T analog front end, but no capacitor-branch current sensor.

II. ENERGY-TRANSFER CAPACITOR MODEL AND HEALTH SIGNATURES

A. Topology, States, and Sign Convention

Consider the classical inverting Ćuk converter in CCM. The state vector is

$$\mathbf{x} = [i_1 \quad i_2 \quad v_C \quad v_o]^T, \quad (1)$$

where $i_1 = i_{L1} > 0$ and $i_2 = i_{L2} > 0$ follow their usual inductor-current directions, v_C is the internal voltage of the ideal element in the energy-transfer capacitor, and $v_o > 0$ is the magnitude of the physically negative output voltage. Let $u \in \{0, 1\}$ denote the main-switch state, with $u = 1$ for the on-state.

The transfer-capacitor current is

$$i_C = (1 - u)i_1 - ui_2, \quad (2)$$

and therefore

$$u = 0 : i_C = i_1, \quad u = 1 : i_C = -i_2. \quad (3)$$

Equation (2) is the topology-specific reconstruction used throughout this work. It avoids a dedicated series current sensor in the capacitor branch, provided that the switching state and both inductor currents are available.

B. Nonideal Capacitor Port

The first-order capacitor model contains an ideal capacitance C in series with ESR r_C :

$$v_T = v_C + r_C i_C, \quad \dot{v}_C = \frac{i_C}{C}, \quad (4)$$

where v_T is the externally measured terminal voltage. Confusing v_T with v_C would attribute the ESR drop to the ideal capacitance and couple the two health indicators.

For numerical conditioning, a fixed nominal capacitance C_b defines

$$\bar{\alpha} = \frac{C_b}{C}, \quad C = \frac{C_b}{\bar{\alpha}}. \quad (5)$$

The health-parameter vector is $\boldsymbol{\theta} = [\bar{\alpha}, r_C]^T$.

TABLE II
PRINCIPAL NOTATION

Symbol	Meaning
u, D, T_s	switch state, duty ratio, switching period
i_1, i_2	input- and output-inductor currents
i_C	reconstructed energy-transfer-capacitor current
v_C, v_T	ideal internal and measured terminal voltages
C, C_b	actual and fixed nominal capacitances
r_C	capacitor ESR at the measurement conditions
$\bar{\alpha}$	normalized inverse capacitance, C_b/C
q	charge transferred inside an accepted window
I_Σ	edge-current sum, $i_1 + i_2$
k_R	training-calibrated gain held fixed during estimation

C. Health Scope

Capacitance loss and ESR growth are treated as condition indicators, not as remaining-useful-life predictions. Because r_C depends on temperature, frequency, operating point, and aging, the method estimates effective ESR under the specified switching and acquisition conditions. The theory assumes recurring valid CCM windows, negligible parameter change within an identification window, and a first-order series C-ESR model. ESL, ringing, DCM, and front-end nonidealities are addressed through safe-window selection, calibration, gating, or explicit limitations.

Only inverse capacitance is normalized: $\bar{\alpha} = C_b/C$ is dimensionless, whereas r_C is estimated directly in ohms. Separate scalar recursions and physical bounds avoid mixing their numerical scales. The predeclared ESR interval is $0.35r_b \leq r_C \leq 2.50r_b$, or 17.5–125 mΩ for $r_b = 50$ mΩ.

III. TOPOLOGY-SYNCHRONOUS DECOUPLED OBSERVATION FRAMEWORK

A. Natural Bidirectional Excitation

At a valid $u : 0 \rightarrow 1$ transition, the capacitor current changes from $i_C^- = i_1^-$ to $i_C^+ = -i_2^+$. Hence

$$\Delta i_C = i_C^+ - i_C^- = -(i_1^- + i_2^+). \quad (6)$$

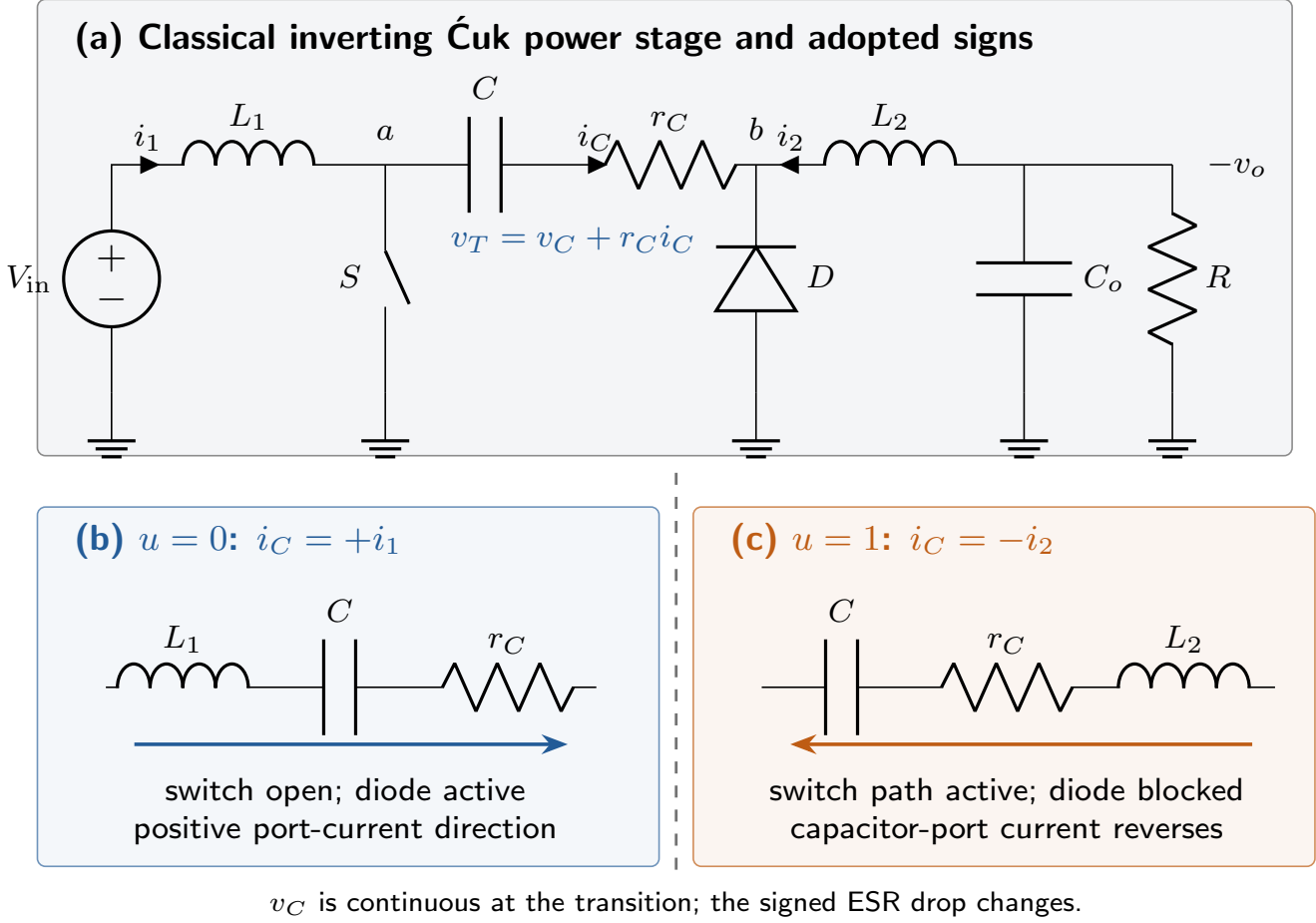


Fig. 1. Classical inverting Ćuk converter and adopted current/voltage signs. The lower panels show the two topologies used by the observer: $u = 0$ gives $i_C = +i_1$, whereas $u = 1$ gives $i_C = -i_2$. The ideal capacitor voltage v_C is continuous at a transition, while the terminal voltage $v_T = v_C + r_C i_C$ contains the ESR step.

TABLE III
CONVERTER AND IDENTIFICATION PARAMETERS

Item	Value
Input voltage	$V_{in} = 24$ V nominal; 19.2/24/28.8 V blind range
Duty ratio	$D = 0.40$ nominal; 0.30/0.40/0.55/0.65 blind range
Switching frequency	$f_s = 50$ kHz ($T_s = 20$ μ s)
Inductors	$L_1 = L_2 = 500$ μ H
Capacitors	$C_b = 100$ μ F; $C_o = 470$ μ F
Nominal ESR/load	$r_b = 50$ m Ω ; $R = 10$ Ω
Primary recursion	$\lambda = 0.9975$; $P_0 = 1000$
Edge calibration	$k_R = 0.9772$; training-calibrated nominal value; case-specific calibrated value used in evaluation
C charge window	$T_{w,C} = 2.0$ μ s
ESR edge-fit window	$T_{w,R} = 2.2$ μ s in the F28379D realization
Health reporting	1024 cycles = 20.48 ms per external report

the edge satisfy

$$v_T^- = v_C + r_C i_1^-, \quad (7)$$

$$v_T^+ = v_C - r_C i_2^+, \quad (8)$$

which gives the ideal ESR signature

$$v_T^- - v_T^+ = r_C (i_1^- + i_2^+). \quad (9)$$

Thus, the topology transition provides a large, signed current change while cancelling the continuous internal voltage.

B. Timestamp-Reconstructed ESR Observation

Direct edge samples are contaminated by acquisition skew, finite bandwidth, and switching transients. The proposed measurement therefore fits local linear models to timestamped points in safe pre- and post-edge windows and extrapolates both models to a common switching timestamp. With reconstructed intercepts $\hat{v}_T^-$ and $\hat{v}_T^+$, the accepted ESR row is

$$z_R = \hat{v}_T^- - \hat{v}_T^+ = k_R I_\Sigma r_C + \nu_R, \quad I_\Sigma = i_1^- + i_2^+, \quad (10)$$

where k_R is a training-locked calibration coefficient and ν_R collects voltage, timestamp, fitting, and residual front-end errors. The reconstruction estimates the discontinuity at

Because the internal ideal-capacitor voltage cannot jump, $v_C^+ = v_C^-$. The terminal voltages immediately before and after

one physical instant instead of subtracting samples taken at different offsets.

Simulation calibration and blind health identification use disjoint cases. Four calibration profiles—nominal, high duty/load, high noise, and nominal parasitic Model B—form the accepted-edge ratio

$$\kappa_m = \frac{z_{R,m}}{I_{\Sigma,m} r_{C,\text{ref},m}}, \quad \hat{k}_R = \text{median}_m \kappa_m, \quad (11)$$

where $r_{C,\text{ref},m}$ is the planted simulation reference ESR. The resulting training-locked coefficient is $k_R = 0.9772$. The manuscript reports k_R to four decimal places; the full-precision value used in all regenerated results is retained in Supplementary Note S2 and the source-traceability record.

The ± 0.006 spread represents case-to-case variation in the calibrated measurement-chain gain. For each simulated case, the realized coefficient is assumed known after calibration and is supplied identically to all compared estimators through $h_R = k_R I_\Sigma$. It therefore tests calibrated unit-to-unit variability rather than residual unknown calibration error; k_R is never updated from blind residuals. A hardware realization would require commissioning with an independently characterized reference capacitor under the final measurement chain.

Thus, k_R is known and fixed within each identification interval; otherwise the edge row identifies only the product $k_R r_C$. The smaller evaluated-envelope value $k_{R,\text{min}}^{\text{eval}} = 0.957654$ is used only for a conservative information bound and does not resolve this scale ambiguity.

C. Safe-Window Charge Observation

For an interval $[t_a, t_b]$ wholly inside one switching state, with the charge-safe-window length denoted by $T_{w,C} = t_b - t_a$, define

$$q = \int_{t_a}^{t_b} i_C(t) dt. \quad (12)$$

Integration of (4) yields

$$\Delta v_T = \frac{q}{C} + r_C \Delta i_C. \quad (13)$$

Using the most recent accepted ESR estimate gives the capacitance pseudo-measurement

$$z_C = \Delta v_T - \hat{r}_C \Delta i_C = \frac{q}{C_b} \bar{\alpha} + \nu_C. \quad (14)$$

The charge row is evaluated away from the switching edge and is capacitance-dominant after ESR compensation. ESR-estimation uncertainty and residual correlation enter the conditional measurement covariance.

More explicitly, first-order substitution of measured end-point current, voltage, charge, and the preceding ESR estimate gives

$$z_C = \frac{q}{C_b} \bar{\alpha} + (r_C - \hat{r}_C) \Delta i_C + \nu_{\Delta v} - \hat{r}_C \nu_{\Delta i} + \nu_q, \quad (15)$$

where ν_q is the voltage-equivalent charge-integration error. In the evaluated implementation its unscaled charge variance is denoted R_q , so that $\text{var}(\nu_q) = \bar{\alpha}^2 R_q$. Treating the preceding

TABLE IV
DECOUPLED OBSERVATION ROWS

Row	Measurement	Regression	Dominant parameter
Edge	$\hat{v}_T^- - \hat{v}_T^+$	$k_R I_\Sigma r_C + \nu_R$	r_C
Charge	$\Delta v_T - \hat{r}_C \Delta i_C$	$(q/C_b) \bar{\alpha} + \nu_C$	$\bar{\alpha}$

ESR error as a zero-mean conditional uncertainty with variance P_{rr} yields

$$R_C \simeq R_{\Delta v} + (\Delta i_C)^2 P_{rr} + \hat{r}_C^2 R_{\Delta i} + \bar{\alpha}^2 R_q. \quad (16)$$

Here $R_{\Delta v} = 2R_v$, $R_{\Delta i} = 2R_i$, and R_q follows from current-noise variance, the ADC interval, and the integration length. Equation (16) propagates uncertainty; a biased $(r_C - \hat{r}_C) \Delta i_C$ remains bounded model mismatch rather than white noise.

D. Validity and Update Policy

An observation is accepted only in CCM with a valid times-tamp, topology state, and complete safe window. Denoting these prechecks by $\mathcal{S}_C$ and $\mathcal{S}_R$, the common evaluation stream implements

$$\begin{aligned} \mathcal{V}_C &= \text{CCM} \wedge \mathcal{S}_C \wedge (n_C > 0), \\ \mathcal{V}_R &= \text{CCM} \wedge \mathcal{S}_R \wedge (I_\Sigma \geq I_{\Sigma,\text{gate}}), \end{aligned} \quad (17)$$

with the predeclared $I_{\Sigma,\text{gate}} = 0.12$ A. Charge acceptance uses the CCM/safe-window flag and a nonzero accepted-charge count; the implementation has no separate numeric q_{gate} . An NIS threshold of 9 may reject additional TS-SLTVKE outliers. A failed gate holds the corresponding estimate. The evaluated-envelope minima in Section IV are proof constants, not online thresholds.

Since the capacitance regressor is $h_C = q/C_b$, low-charge rows inherently contribute little information. The positive information guarantee is therefore restricted to the evaluated valid-CCM envelope rather than asserted for arbitrarily small q .

IV. FINITE-WINDOW IDENTIFIABILITY AND INFORMATION BOUNDS

A. Information Structure

For accepted scalar rows, define $h_C = q/C_b$ and $h_R = k_R I_\Sigma$, with conditional variances R_C and R_R . Because the two observations act along disjoint parameter directions, the finite-window information Gramian is

$$\mathbf{G}_{\theta,N} = \sum_{k \in \mathcal{K}_C} \frac{1}{R_{C,k}} \begin{bmatrix} h_{C,k}^2 & 0 \\ 0 & 0 \end{bmatrix} + \sum_{k \in \mathcal{K}_R} \frac{1}{R_{R,k}} \begin{bmatrix} 0 & 0 \\ 0 & h_{R,k}^2 \end{bmatrix}. \quad (18)$$

Suppose every accepted charge interval has length at least $T_{w,C,\text{min}}$ and satisfies $|i_C| \geq I_{C,\text{min}}^{\text{eval}}$ without changing sign. Then $|q| \geq q_{\text{min}}^{\text{eval}} = I_{C,\text{min}}^{\text{eval}} T_{w,C,\text{min}}$. If $R_C \leq R_{C,\text{max}}$, $R_R \leq R_{R,\text{max}}$, the calibrated k_R remains fixed with known conservative evaluated-envelope lower bound $k_{R,\text{min}}^{\text{eval}} > 0$, and

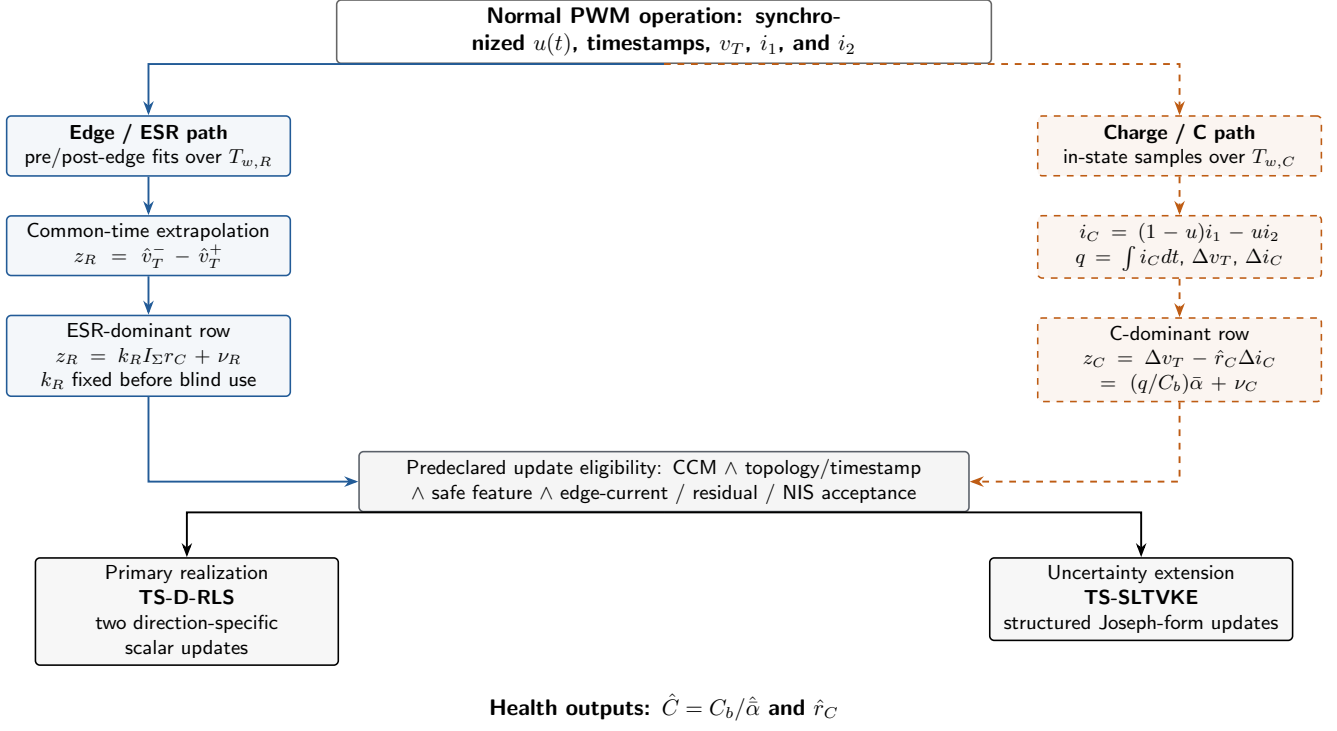


Fig. 2. Core topology-synchronous decoupled observation framework. The edge-fit window $T_{w,R}$ and charge-safe window $T_{w,C}$ remain distinct; the downstream recursive estimators use standard scalar update principles.

$I_{\Sigma} \geq I_{\Sigma, \min}^{\text{eval}} > 0$, windows containing m_C charge rows and m_R edge rows have the lower bounds

$$\underline{\mu}_C = m_C \frac{(I_{C, \min}^{\text{eval}} T_{w, C, \min} / C_b)^2}{R_{C, \max}}, \quad (19)$$

$$\underline{\mu}_R = m_R \frac{(k_{R, \min}^{\text{eval}})^2 (I_{\Sigma, \min}^{\text{eval}})^2}{R_{R, \max}}. \quad (20)$$

Proposition 1 (Operating-condition-induced finite-window identifiability): Assume valid CCM; k_R known and fixed during each identification window; sign-invariant accepted charge windows; positive accepted edge-current sum; finite conditional covariances; and at least one accepted charge and edge row in a finite window. Then

$$\mathbf{G}_{\theta, N} \succeq \text{diag}(\underline{\mu}_C, \underline{\mu}_R) \succ \mathbf{0}, \quad (21)$$

and (C, r_C) is locally identifiable over the accepted finite window.

Proof: Each charge row contributes a positive semidefinite matrix only in the $\bar{\alpha}$ direction, and each edge row contributes one only in the r_C direction. Applying the physical lower bounds in (19) and (20), summing m_C and m_R accepted contributions, and noting that both lower bounds are strictly positive establishes (21). Since $C = C_b/\bar{\alpha}$ is locally one-to-one for $\bar{\alpha} > 0$, local identifiability transfers from $(\bar{\alpha}, r_C)$ to (C, r_C) . $\square$

The result is conditional: PWM alone does not guarantee persistent excitation at every load or conduction mode. The Čuk topology supplies the physical directions, while accepted rows certify when the bound applies. The superscript “eval”

marks minima over the evaluated CCM envelope; these constants are neither online gates nor post-hoc tuning values.

B. Numerical Lower-Bound Check

Across 45 independent Model-B CCM cases, the weakest physical quantities were

$$I_{C, \min}^{\text{eval}} = 0.330108 \text{ A}, \quad I_{\Sigma, \min}^{\text{eval}} = 0.438504 \text{ A}, \\ q_{\min}^{\text{eval}} = 0.660217 \mu\text{C}. \quad (22)$$

with $T_{w, C, \min} = 2.0 \mu\text{s}$ and $k_{R, \min}^{\text{eval}} = 0.957654$. The resulting weakest information bounds were

$$\underline{\mu}_C = 0.217943, \quad \underline{\mu}_R = 880.008. \quad (23)$$

The minimum ratios of empirical information to the corresponding conservative bound were 1.02872 and 1.02334. Hence every retained case exceeded its computed bound, while the weakest case rather than a nominal case determines the reported constants.

C. Covariance Contraction for the Uncertainty Extension

For the uncertainty extension, let P_n be a posterior variance, Q_N the process variance accumulated over a finite window, and $\mu \geq \underline{\mu} > 0$ the accepted information. Then

$$P_{n+1} \leq \frac{P_n + Q_N}{1 + \mu(P_n + Q_N)}. \quad (24)$$

Proposition 2 (Finite-window covariance boundedness): For recurring valid updates, finite $0 \leq Q_N \leq \bar{Q}_N$, and $\mu \geq \underline{\mu} > 0$,

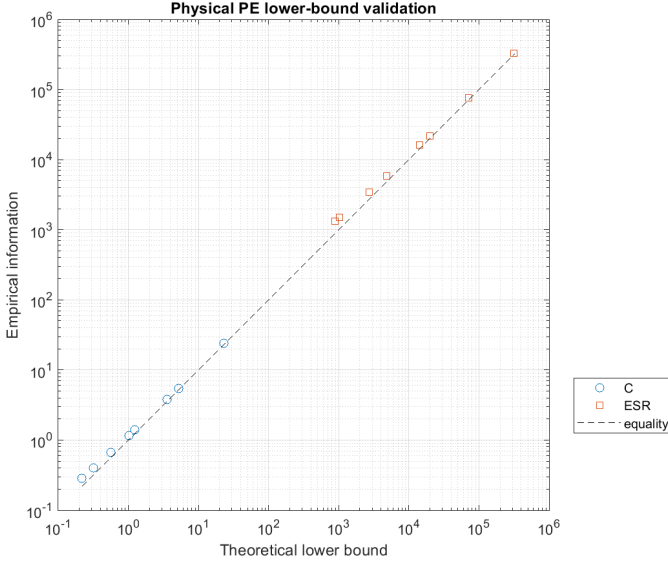


Fig. 3. Theoretical lower bounds versus empirical finite-window information over the Model-B CCM cases. Circles denote normalized inverse capacitance and squares denote physical ESR; their parameterization-dependent magnitudes and units are not directly comparable. All accepted cases lie above the equality line.

the posterior covariance satisfies (24). For constant Q_N , its positive fixed point is

$$P^* = \frac{-Q_N + \sqrt{Q_N^2 + 4Q_N/\mu}}{2}. \quad (25)$$

For $Q_N = 0$, $P_n^{-1} \geq P_0^{-1} + n\mu$.

This conditional linear-model result is consistent with standard filtering theory [23], [24]; it is not a global nonlinear-converter convergence theorem. Appendix B gives the derivation and validated fixed-point values. Projection was inactive in 200 legal-PE seeds and is treated as a safety constraint rather than the source of contraction.

V. ONLINE ESTIMATION REALIZATIONS

Both realizations consume the same accepted edge and charge rows from Section III, separating observation effects from recursive-kernel effects.

A. Primary Realization: TS-D-RLS

The primary implementation uses one scalar RLS recursion for $\bar{\alpha}$ and one for r_C . For direction $j \in \{C, R\}$, write an accepted row as $z_{j,k} = h_{j,k}\theta_{j,k} + \nu_{j,k}$. With forgetting factor $\lambda = 0.9975$,

$$K_{j,k} = \frac{P_{j,k-1}h_{j,k}}{\lambda + h_{j,k}^2 P_{j,k-1}}, \quad (26)$$

$$\hat{\theta}_{j,k} = \hat{\theta}_{j,k-1} + K_{j,k}(z_{j,k} - h_{j,k}\hat{\theta}_{j,k-1}), \quad (27)$$

$$P_{j,k} = \lambda^{-1}(P_{j,k-1} - K_{j,k}h_{j,k}P_{j,k-1}). \quad (28)$$

Only the direction associated with an accepted observation is updated. Validity gating rejects observations that fail pre-declared edge/charge acceptance rules; projection enforces physical parameter intervals. The reported health values are

$$\hat{C} = \frac{C_b}{\hat{\alpha}}, \quad \widehat{\text{ESR}} = \hat{r}_C. \quad (29)$$

Equations (26)–(28) are standard RLS recursions. The TS-D-RLS name denotes their topology-synchronous, direction-specific use with the proposed observations; RLS itself is not presented as the novelty.

The implementation reconstructs i_C , validates CCM, forms edge or charge features, applies physical/residual gates, and updates only the active scalar direction. Internal updates are consolidated into an external health report every 1024 switching cycles.

B. Uncertainty-Aware Extension: TS-SLTVKE

The extension maintains the structured state

$$\mathbf{x}_k = [v_{C,k} \quad \bar{\alpha}_k \quad r_{C,k}]^T. \quad (30)$$

For an intersample charge increment q_k , the ideal propagation contains

$$\mathbf{F}_k = \begin{bmatrix} 1 & q_k/C_b & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (31)$$

The stable terminal-voltage, charge, and edge rows use the structured directions

$$\mathbf{H}_V = [1, 0, 0], \quad \mathbf{H}_C = [0, q/C_b, 0], \quad \mathbf{H}_R = [0, 0, k_R I_\Sigma]. \quad (32)$$

Accepted rows use scalar Joseph-form covariance updates, predeclared process and measurement covariances, and normalized-innovation gating. This extension provides posterior-covariance-based confidence reporting; it is not assumed to lead every point-estimation metric, and its negative dynamic results are retained in Section VII.

The covariance readout is intrinsic to TS-SLTVKE. The TS-D-RLS interval is instead an auxiliary residual-weighted sandwich diagnostic that does not affect the estimate; the two intervals therefore have different statistical meanings.

VI. VERIFICATION METHODOLOGY

A. Model and Data Hierarchy

The evaluation separates equation checks, independent plant traces, estimator comparisons, and device-realistic acquisition:

- 1) Model A is a switched-equation Ćuk model for sign, timing, and rapid dynamic checks.
- 2) Model B is an independently constructed Simscape model; 45 traces establish current scale, edge slope, and the blind observation data.
- 3) The estimator comparison uses 48 stratified blind operating/health cases, four noise profiles, and four residual timing-skew levels under common O1 observations.
- 4) The F28379D model adds register-realizable sampling, aperture/occupancy, reference/clock effects, AFE residuals, and Monte Carlo variation; it remains a simulation.

B. Factorial and Blind Comparisons

The observation–estimator factorial crosses a conventional mixed observation set (O0) and the proposed direction-decoupled set (O1) with RLS (E1), Dual EKF (E2), and the structured LTV/Joseph estimator (E3). All $2 \times 3 \times 768 = 4608$

paired rows are retained. The final comparison evaluates TS-D-RLS, TS-SLTVKE, and Dual EKF under O1.

The broader baseline study includes closed-form, mixed-RLS, augmented/Dual-EKF, and Ćuk-adapted wavelet-KF methods. An NPC spectral method was excluded because the literature provides no unique mapping to the Ćuk edge/charge convention; inventing one would not reproduce the published baseline.

C. Fairness Rules and Metrics

All methods use identical blind cases, observations, seeds, initialization distributions, noise/timing realizations, and pre-declared hyperparameters. Failed or saturated rows are retained, no method is retuned after blind evaluation, and paired effects use 10,000 bootstrap resamples.

Abrupt C/ESR discontinuities are estimator stress tests. The principal dynamic cases are linear and smooth C-down, ESR-up, and joint ramps lasting 0.1, 1, 10, and 100 s. Long ramps use trace-derived O1 sufficient statistics, while 0.1 s cases are cross-checked at every switching pair with Model A. Load, input-voltage, and duty transitions hold health constant to measure false-health response. None of these trajectories is hardware aging data.

Static endpoint errors are

$$e_C = 100 \left| \hat{C}_N / C - 1 \right|, \quad e_R = 100 \left| \hat{r}_{C,N} / r_C - 1 \right|, \quad (33)$$

with MAPE and p95 computed over retained rows. “Convergence cycles” is the first cycle starting 32 consecutive cycles with both relative errors no greater than 5%; a miss receives the 1024-cycle horizon. Rejected cycles hold the estimate and remain in the count. This internal settling metric differs from the 1024-cycle (20.48-ms) external reporting cadence.

For a trajectory $y(t)$ with initial truth y_0 , the implemented tracking metric is

$$\text{nRMSE}_y = \sqrt{\frac{\int (\hat{y} - y)^2 dt}{\int (y - y_0)^2 dt}}. \quad (34)$$

Combined nRMSE is the unweighted mean of finite C and ESR values. Normalized lag is the largest finite C/ESR delay from zero-mean cross-correlation and 50%-trajectory crossings, divided by trajectory duration. Missed threshold crossings are retained but are not used to select the primary estimator. Supplementary Note S3 records the complete threshold and bootstrap definitions.

VII. RESULTS

A. Attribution to the Observation Framework

The observation–estimator factorial provides the clearest attribution. Replacing O0 with O1 while retaining the E1 RLS kernel reduced mean capacitance MAPE from 7.0366% to 0.3727%, mean ESR MAPE from 1.1970% to 0.2233%, and mean internal convergence from 665.41 to 13.21 switching cycles. The paired ESR reduction was 0.97369 percentage points (95% bootstrap interval 0.92352–1.0265); all 4608 rows were retained. Supplementary Note S3 gives the other paired intervals.

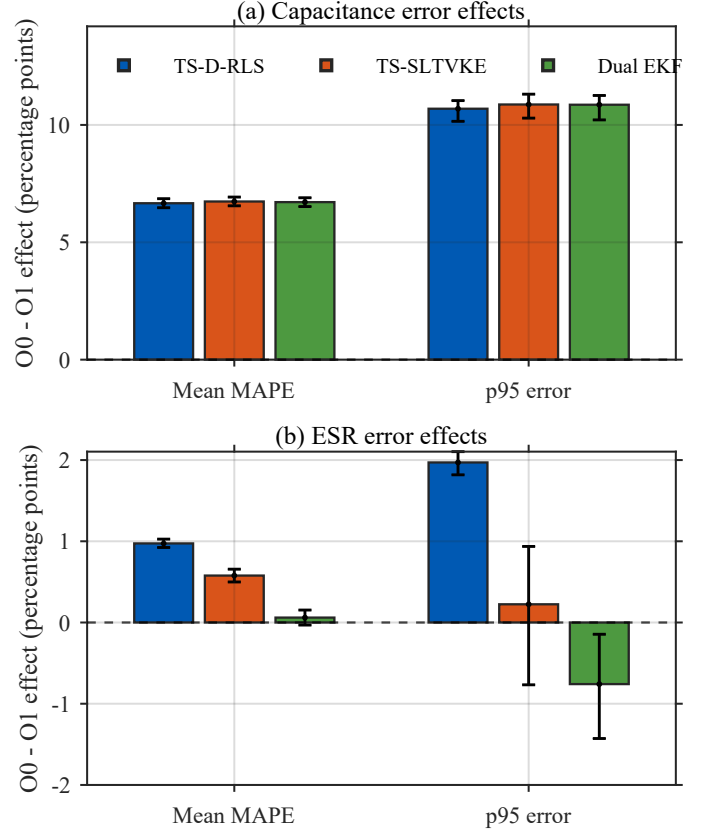


Fig. 4. Paired O0–O1 effects on MAPE and p95 error with bootstrap intervals. Positive values favor the topology-decoupled O1 observations; the adverse Dual-EKF ESR-p95 effect is retained.

O1 also reduced capacitance MAPE by more than 6.7 percentage points for Dual EKF and E3 LTV/Joseph. The improvement was not universal: the Dual-EKF ESR-p95 effect was negative and its interval did not favor O1. Thus, the factorial supports the observation framework without implying monotonic improvement in every estimator metric.

B. Static Estimator Comparison Under O1

Table VI compares the three candidates on the $48 \times 4 \times 4$ O1 matrix. TS-SLTVKE and Dual EKF gave slightly lower mean capacitance error, whereas TS-D-RLS gave the lowest ESR error and fastest native convergence; none diverged. TS-D-RLS is therefore adopted as the primary lightweight realization because it provides the more favorable aggregate tradeoff among ESR accuracy, convergence, degradation tracking, and arithmetic cost over the evaluated cases.

C. Noise and Timing Robustness

All combinations of four noise profiles and 0, 20, 50, and 100 ns residual mismatch were retained. At 50 ns under F28379D-realistic noise, C/ESR p95 errors were 0.4964%/0.3385% for TS-D-RLS, 0.4132%/0.6888% for TS-SLTVKE, and 0.4211%/1.9141% for Dual EKF. Figure 6 visualizes that device-realistic profile; C and ESR percentiles are shown separately because their arithmetic mean has no independent interpretation.

TABLE V
BLIND VERIFICATION AND ESTIMATOR SETTINGS

Item	Predeclared setting	Role/source boundary
Health/load matrix	$C/C_b \in \{0.8, 0.9, 1.0\}$; $r_C/r_b \in \{1, 1.5, 2\}$; load factors 0.58/1/1.45	Stratified blind plant matrix; retained CCM rows only
Initialization	realized $\hat{C}_0/C \in [0.708, 1.287]$; $\hat{r}_{C,0}/r_C \in [0.512, 1.483]$	Read from the blind-case table
Parameter bounds	$C/C_b \in [0.65, 1.35]$; $r_C/r_b \in [0.35, 2.50]$	Predeclared projection limits; failed/saturated rows retained
Noise profiles	$(\sigma_v, \sigma_i) = (1 \text{ mV}, 0.5 \text{ mA}), (5 \text{ mV}, 2 \text{ mA}), (10 \text{ mV}, 5 \text{ mA}), \text{ and } (2.2 \text{ mV}, 1.2 \text{ mA})$	Shared realization; last pair is F28379D-realistic
Residual skew	0, 20, 50, and 100 ns	Shared by all compared estimators
Online validity/gating	valid CCM/state/time and safe feature; $I_{\Sigma, \text{gate}} = 0.12 \text{ A}$; no numeric q_{gate} ; TS-SLTVKE NIS gate 9	Fixed before evaluation, not inferred from PE minima
Evaluated PE constants	$I_{C, \min}^{\text{eval}} = 0.330108 \text{ A}$; $I_{\Sigma, \min}^{\text{eval}} = 0.438504 \text{ A}$; $q_{\min}^{\text{eval}} = 0.660217 \mu\text{C}$	Minima over 45 evaluated Model-B CCM cases; proof only, never online tuning thresholds
Blind comparison	48 cases $\times$ 4 noise levels $\times$ 4 skews; O0/O1 $\times$ RLS/Dual EKF/LTV-Joseph	O1 is then shared by TS-D-RLS, TS-SLTVKE, and Dual EKF
Statistics/dynamics	10,000 paired bootstrap resamples; 0.1/1/10/100-s C, ESR, and joint ramps	Load, input-voltage, and duty steps hold physical health fixed
TS-SLTVKE settings	$Q_\alpha = 2 \times 10^{-9}$, $Q_r = 5 \times 10^{-9}$, $P_{0, \alpha} = 0.0144$, $P_{0, r}$ scale 0.45, NIS 9	Held fixed before blind comparison
Device acquisition	1.093 MS/s/module; 0.5- μs guard; $T_{w, R} = 2.2 \mu\text{s}$; three points/side	Constrained F28379D simulation only

TABLE VI
STATIC O1 ESTIMATOR COMPARISON WITH INTERNAL CONVERGENCE

Method	C MAPE (%)	ESR MAPE (%)	C/ESR p95 (%)	Conv. (int. cycles)
TS-D-RLS	0.3727	0.2233	1.3182/0.7002	13.21
TS-SLTVKE	0.3011	0.6446	1.1069/2.4992	33.09
Dual EKF	0.3043	1.1938	1.1492/3.8602	40.58

TABLE VII
AGGREGATE SLOW-DEGRADATION TRACKING

Ramp (s)	Combined nRMSE		Normalized lag (%)	
	TS-D-RLS	TS-SLTVKE	TS-D-RLS	TS-SLTVKE
0.1	0.1161	0.2160	8.053	22.027
1	0.0163	0.0454	1.360	3.537
10	0.0160	0.0456	1.752	4.492
100	0.0213	0.0477	2.827	4.868

D. Abrupt Stress Tests

Abrupt discontinuities are recovery stress tests, not aging events. TS-SLTVKE did not satisfy the convergence criterion within 257 post-step cycles for either health step; its tail errors were 19.2911% after $C \rightarrow 0.8C$ and 49.9384% after $r_C \rightarrow 2r_C$. These retained negative results are a principal reason it is presented as an uncertainty extension rather than the sole estimator.

E. Slow Degradation Tracking

Across the 0.1–100 s sweep in Table VII, TS-D-RLS had lower aggregate combined nRMSE and normalized lag at all four durations. Individual trajectories can still favor TS-SLTVKE; one 10-s C-ramp case did so. The supported conclusion is therefore aggregate rather than universal.

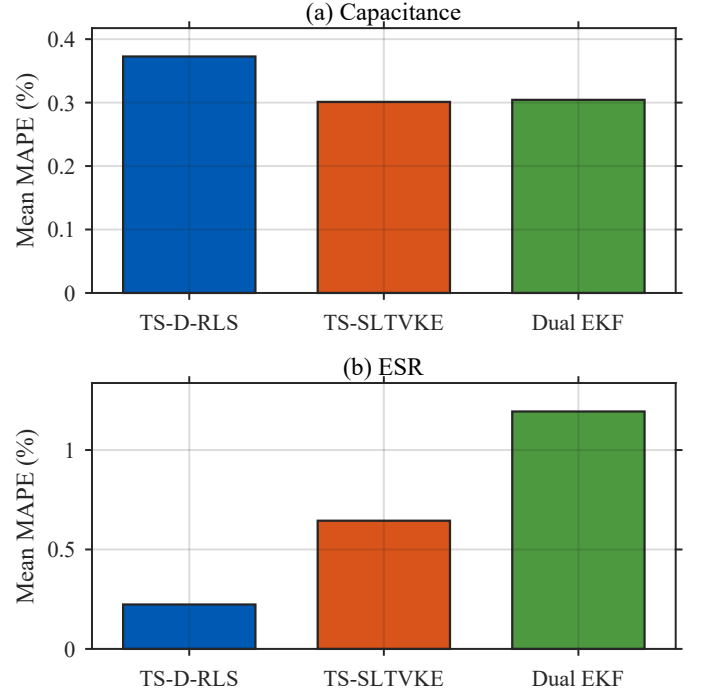


Fig. 5. Static capacitance and ESR errors under common O1 observations. The tradeoff motivates the dual-realization decision.

F. Operating-Point Transient Immunity

Load, input-voltage, and duty transitions held health fixed. TS-SLTVKE reduced the mean false-C peak (0.7570% versus 1.3017%), whereas TS-D-RLS reduced the mean false-ESR peak (1.8474% versus 5.1919%) and recovered faster. Neither method therefore dominates operating-transient immunity; Supplementary Fig. S4 shows the retained load-step traces.

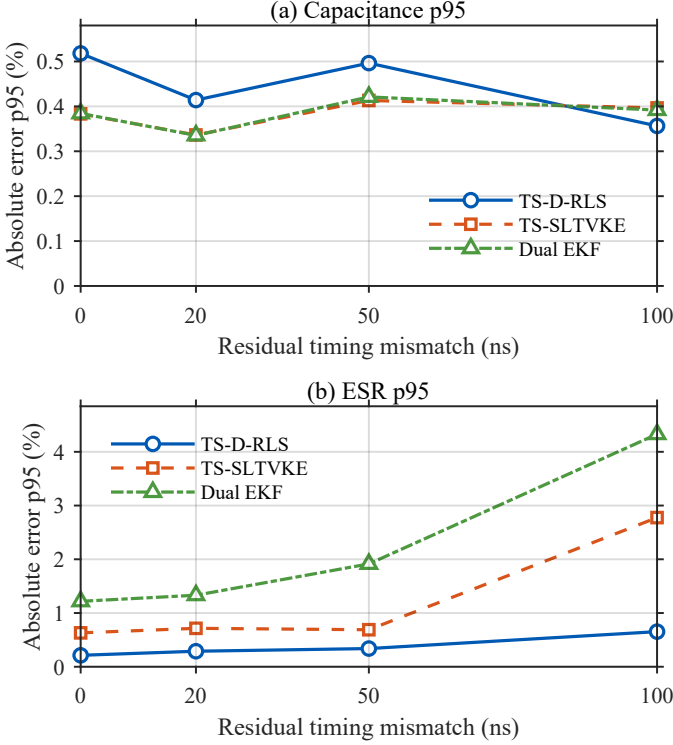


Fig. 6. C and ESR p95 errors versus residual timing mismatch under the F28379D-realistic noise profile and shared O1 observations. Distinct line and marker styles preserve method identity in grayscale.

TABLE VIII
PER-ACCEPTED-OBSERVATION COMPLEXITY

Method	Mult.	Add.	Div.	Est. time (μ s)
TS-D-RLS	28	20	1	1.36
TS-SLTVKE	46	34	1	2.12
Dual EKF	62	47	2	3.05

G. Complexity and Uncertainty

On the common arithmetic basis in Table VIII, TS-D-RLS, TS-SLTVKE, and Dual EKF require 28, 46, and 62 multiplications per accepted observation. Their joint 95% interval coverages are 13.00%, 88.93%, and 81.90%, respectively. Thus, TS-D-RLS provides the lower-cost tracking realization. TS-SLTVKE is retained as an uncertainty-aware realization of the same observation framework because its interval coverage is substantially better calibrated, despite less favorable aggregate tracking and point-estimation tradeoffs. The listed times are arithmetic estimates, not measured target WCET; acquisition and feature-processing costs are separate.

VIII. F28379D DEVICE-REALISTIC FEASIBILITY

The device-realistic study tests whether the internal ADC architecture can acquire the proposed observations under register-realizable timing and an explicit high-common-mode analog front end (AFE). Four synchronized 16-bit differential ADC modules use a 320-ns aperture and sustain 1.093 MS/s per module. A one-edge schedule places three samples on each side of the reconstructed edge within $T_{w,R} = 2.2 \mu$ s after a

TABLE IX
F28379D DEVICE-REALISTIC SUMMARY

Item	Specified value
ADC architecture	four modules, 16-bit differential
Aperture/rate	320 ns / 1.093 MS/s per module
Edge acquisition	0.5- μ s guard; $T_{w,R} = 2.2 \mu$ s; 3 points/side
AFE boundary	differential path; edge-band CMRR ≥ 70 dB
Health reporting	1024 cycles (20.48 ms)
Worst p95 C/ESR	1.6425% / 2.6109%

0.5- μ s guard. Direct ADC connection is invalid over the simulated common-mode range, so feasibility requires the specified differential AFE, edge-band common-mode rejection, external reference, clock, and calibration residual.

Seven representative conditions with 200 Monte Carlo seeds each yielded worst-case p95 capacitance and ESR errors of 1.6425% and 2.6109%; every condition met its predeclared threshold. Table IX summarizes the evidence needed for the feasibility claim. Register timing, aperture margin, DMA traffic, utilization, AFE component assumptions, common-mode extrema, and coverage ranges are retained in Supplementary Note S1.

The 1.36 μ s TS-D-RLS value is an estimator-arithmetic estimate, not platform WCET. Firmware compilation, linker placement, measured execution time, and bench AFE verification remain open; the result supports a constrained simulation architecture rather than hardware validation.

IX. DISCUSSION

A. Why Observation Design Matters

The O0–O1 factorial separates observation design from estimator complexity. With the RLS kernel unchanged, O1 reduced capacitance MAPE from 7.0366% to 0.3727%, ESR MAPE from 1.1970% to 0.2233%, and mean internal convergence from 665.41 to 13.21 cycles. This attribution agrees with the circuit physics: the switching edge exposes the ESR drop, while the in-state charge balance exposes inverse capacitance. More elaborate recursion cannot reliably compensate for observations that mix these effects or compare mismatched timestamps.

The scoped literature matrix contains no disclosed method combining the Čuk transfer-capacitor target, $+i_{L1} \leftrightarrow -i_{L2}$ commutation, timestamp-reconstructed edge information, and direction-specific charge/edge updates. This is a bounded review finding, not a universal first-in-literature claim.

B. Why TS-D-RLS Is Primary

TS-D-RLS is primary because it combines low ESR error, rapid convergence, low operation count, and the better aggregate ramp-tracking tradeoff. This choice neither makes RLS a novelty nor implies superiority in every metric. TS-SLTVKE remains valuable for confidence-aware reporting: its joint coverage is much closer to nominal, but its slower ramp response and retained abrupt-step failures limit it as the sole point estimator. An adaptive change detector or process-noise policy could alter that tradeoff but lies outside the evaluated design.

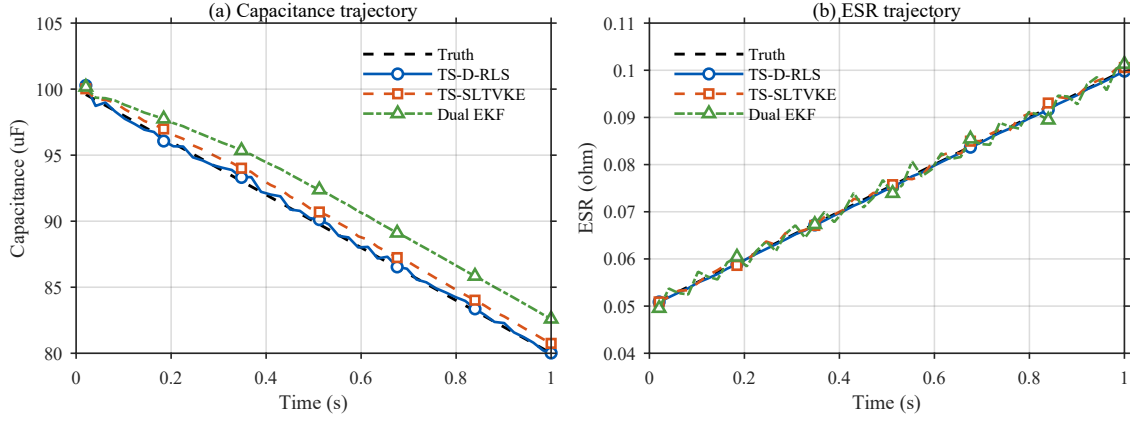


Fig. 7. Representative joint C-down/ESR-up health ramps. Long-duration trajectories use trace-derived O1 sufficient statistics rather than full Simscape transients or measured aging data.

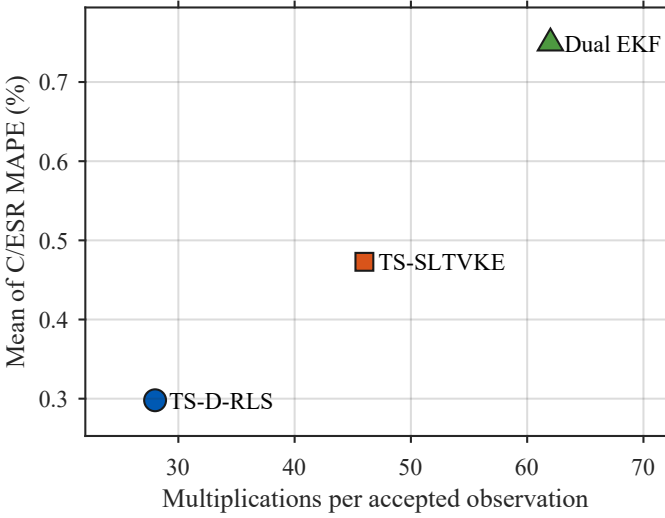


Fig. 8. Accuracy-complexity tradeoff under the common operation-count model. The ordinate is the arithmetic mean of C and ESR MAPE values, used only as a visual Pareto summary and not as the estimator-selection objective.

C. Limitations

The evidence has six principal boundaries.

- 1) Identifiability and updates require valid CCM; DCM and weak-excitation intervals hold the estimate. The method assumes availability of both inductor currents; applications lacking either measurement require an additional sensor or independently validated current reconstruction.
- 2) The reported ESR is an effective condition indicator at the modeled switching, thermal, and frequency conditions, not a temperature-independent aging variable.
- 3) The 0.1–100 s trajectories are synthetic or trace-derived bandwidth tests, not run-to-failure data; abrupt discontinuities are recovery stress tests rather than aging events.
- 4) Hardware validation remains open for edge ringing, isolation/probing, AFE common-mode rejection, and production calibration.
- 5) The F28379D study is device-realistic simulation; target compilation and measured WCET have not been demonstrated.

- 6) The auxiliary TS-D-RLS intervals are under-covered, while TS-SLTVKE retains the reported dynamic limitations; confidence calibration and point-tracking performance therefore remain a tradeoff.

D. Experimental Roadmap

The next stage should use a controlled C/ESR network in the energy-transfer branch, the specified F28379D schedule, separate absolute- and edge-voltage paths, and synchronized inductor-current measurements. Tests should sweep input voltage, duty ratio, load, and temperature; characterize ringing and AFE rejection; verify calibration against offline impedance measurements; and measure WCET before firmware release.

X. CONCLUSION

This paper converted the normal bidirectional energy transfer of a Ćuk converter into two topology-synchronous capacitor-health observations. A timestamp-reconstructed switching-edge row provides ESR-dominant information, while an ESR-compensated safe-window charge row provides capacitance-dominant information. Under valid CCM gates, positive finite-window information bounds establish local C–ESR identifiability and state the operating conditions under which the separation is informative.

On the blind matrix, the primary TS-D-RLS realization achieved 0.3727% mean capacitance MAPE and 0.2233% mean ESR MAPE. It also provided the better aggregate ramp-tracking tradeoff and the lowest operation count among the final candidates. TS-SLTVKE remains an uncertainty-aware extension with better confidence calibration but retained negative dynamic results. A constrained F28379D simulation supported device-realistic acquisition feasibility with worst p95 C/ESR errors of 1.6425%/2.6109%.

The main conclusion concerns observation design rather than estimator complexity. Hardware and accelerated-aging studies must still validate AFE common-mode behavior, ringing, temperature dependence, calibration transfer, and measured execution time before experimental or lifetime claims are warranted.

APPENDIX A SWITCHED EQUATIONS UNDER THE ADOPTED SIGN CONVENTION

For completeness, the first-order switched equations used to verify the observation signs retain the transfer-capacitor ESR and set the other parasitics to zero. Here v_C is the ideal capacitor voltage and $v_T = v_C + r_C i_C$. With $u = 1$ (main switch on),

$$\dot{i}_1 = V_{in}/L_1, \quad \dot{i}_2 = (v_C - r_C i_2 - v_o)/L_2, \quad (35)$$

$$\dot{v}_C = -i_2/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (36)$$

With $u = 0$ (main switch off),

$$\dot{i}_1 = (V_{in} - v_C - r_C i_1)/L_1, \quad \dot{i}_2 = -v_o/L_2, \quad (37)$$

$$\dot{v}_C = i_1/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (38)$$

These equations yield (2) and match the specified Model-A implementation after its optional inductor, switch, diode, and source parasitics are set to zero. The ESR signs differ between states because $i_C = +i_1$ for $u = 0$ and $i_C = -i_2$ for $u = 1$.

APPENDIX B SCALAR COVARIANCE ENVELOPE

Let the posterior scalar variance at the start of a finite window be P_n . Process accumulation gives $P_n^- \leq P_n + Q_N$. Assimilating total scalar information μ gives

$$P_{n+1} \leq \left(\frac{1}{P_n + Q_N} + \mu \right)^{-1}, \quad (39)$$

which is (24). Setting $P_{n+1} = P_n = P^*$ gives

$$\mu(P^*)^2 + \mu Q_N P^* - Q_N = 0, \quad (40)$$

whose positive root is (25). The map is increasing and bounded above by $1/\mu$. When $Q_N = 0$, inversion gives $P_{n+1}^{-1} \geq P_n^{-1} + \mu$ and hence $P_n \leq (P_0^{-1} + n\mu)^{-1}$. With bounded conditional mismatch, a finite mismatch-dependent error term must be added; no universal numerical value is assigned because the nonlinear remainder is case dependent.

The largest validated fixed points were 0.0112191 in the normalized-capacitance direction and 8.82437×10^{-6} in the ESR direction.

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